ADITYA UNIVERSITY

Worksheet - 01

Course: Aptitude **Topic**: Number System

1. What is the difference between the square of 15 and the square of 10?
   A) 100 **B) 125** C) 150 D) 175
   Sol: $15^2 = 225$ and $10^2 = 100$, so the difference is 225 - 100 = 125.

2. What is the sum of the first 5 even numbers?
   A) 20 B) 25 **C) 30** D) 35
   Sol: The first 5 even numbers are 2, 4, 6, 8, and 10. Their sum is 2+4+6+8+10 = 30

3. How many Composite Numbers in Between 1-100?
   A) 30 B) 25 C) 75 **D) 74**
   Sol: Except 1 and prime numbers (25+1)=100-26=74

4. How many digits are in the number 2^10?
   A) 3 B) 6 C) 5 **D) 4**
   Sol: 2^10 = 1024, which has 4 digits.

5. What is the last digit of 7^4?
   A) 3 B) 7 **C) 1** D) 9
   Sol: 7^4 = 2401, so the last digit is 1.

6. Find the sum of the digits of 2^10.
   A) 7 B) 6 C) 8 D) 9
   Sol: 2^10 = 1024; the sum of its digits is 1+0+2+4 = 7.

7. If a number is divisible by both 8 and 12, what is the smallest such number?
   A) 16 **B) 24** C) 32 D) 48
   Sol: The least common multiple of 8 and 12 is 24.

8. What is the remainder when 12345 is divided by 8?
   A) 3 **B) 1** C) 4 D) 2
   Sol: Using Divisibility rule of 8 i.e A number is divisible by 8 if the last three digits of the number form a number that is divisible by 8.

9. What is the sum of the first 7 prime numbers?
   A) 58 B) 56 C) 60 D) 62
   Sol: The first 7 prime numbers are 2, 3, 5, 7, 11, 13, 17, which sum to 58.

10. What is the product of the digits of 567?
    A) 180 **B) 210** C) 240 D) 300
    Sol: The product of the digits of 567 is 5 × 6 × 7 = 210.

11. A number 12345*7 divided by 9.Then find the value of *
    A) 1 B) 2 **C) 5** D) 9

Sol: 1+2+3+4+5+*+7=divisible by 9==>22+*=27==>*=5

12. Find the **smallest 3-digit number** that leaves remainder **1** when divided by **2, 3, 4, 5 and 6**.
A) 101 B) 119 **C) 121** D) 181
Sol:
If a number leaves remainder 1, then **number – 1** is divisible by all divisors.
LCM(2,3,4,5,6) = 60.
So number = 60k + 1.
Smallest 3-digit value = **121**.

13. What is the **smallest number** that must be multiplied with **1470** to make it a **perfect square**?
A) 15 B) 20 **C) 30** D) 60
Sol:
$1470 = 2 \times 3 \times 5 \times 7^2$.
For a perfect square, all powers must be even.
Missing factors: 2, 3, 5 → product = **30**.

14. Which of the following is **always divisible by 9** for any 4-digit number?
A) Number − digit sum B) Number − reverse C) Digit sum **D) Both A and B**
Sol:
A number and its digit sum leave the **same remainder when divided by 9**. The difference between a number and its reverse is also divisible by 9.

15. Find the unit digit of 1463^92.
A) 3 B) 9 C) 7 **D) 1**
Sol:
The power cycle pattern for 3 is 3, 9, 7, 1.
(92/4) gives the remainder 0(Since, the power cycle of 8 repeats for every 4 values , the power is being divided by 4.)
If the remainder is 0, then the 4th value in the power cycle will be the unit digit.
Hence, the last digit is 1.

16. The sum of **five consecutive integers** is **S**. What is the sum of the **next five consecutive integers**?
A) S + 15 B) S + 20 **C) S + 25** D) S + 30
Sol:
Each number increases by 5.
Total increase = 5 numbers × 5 = **25**.

17. Find the remainder when $2^{100}$is divided by 15.
A) 0 **B) 1** C) 2 D) 100
Sol:
2^100/15=(2^4)^25/(2^4-1)=(1)^25=1

18. What is the remainder when $3^{40}$is divided by 80?
A) 9 B) 27 **C) 1** D) 49
Sol: 3^40/80=(3^4)^10/(3^4-1)=(1)^10=1

19. How many **positive integer solutions** satisfy $5x + 7y = 108$?
A) 2 **B) 3** C) 4 D) 5
Sol:
Solving the linear equation gives valid solutions:
(2,14), (9,9), (16,4).
Total = **3**.

20. 20.Find the **smallest positive integer** leaving remainders **2, 3 and 2** when divided by **3, 5 and 7** respectively. **(CTS)**
A) 17 B) 23 C) 53 D) 65
Sol: Go Through Options

21. Find the number of factors of 72900. **(TCS)**
A) 68 **B) 63** C) 54 D) 66
Sol:72900= 729 x 100 => 2^2*3^6*5^2
If n is the number and n=(x^a*y^b*z^c) then,
Number of factors = (a+1) (b+1) (c+1)
Number of factors of 72900 = (2+1)*(6+1)*(2+1)
= 3 * 7* 3
= 63

22. In how many ways 124 can be written as product of two numbers? **(TCS)**
A) 6 B) 12 C) 24 **D) 3**
Sol: Product of two numbers = No of factors / 2
Factors of 124 = 4*31 = 22 * 31 No of factors = (2+1)*(1+1) = 3*2 = 6
Product of two numbers = 6/2 = 3 ways

23. Find The total number of Factors Of 240. **(Deloitte)**
A) 26 B) 24 **C) 20** D) 28
Sol: 240=2^4*3^1*5^1==>To find total number of factors=(4+1)(1+1)(1+1)=5*2*2=20

24. Find the rightmost non-zero integer of the expression 2840^125 + 2870^155. **(CTS)**
A) 6 **B) 4** C)2 D) 8
Sol: Considering from the right end towards left,
2870^155 will have 155 zeroes and then we will get a rightmost non-zero integer.
2840^125 will have 125 zeroes and then a rightmost non-zero integer. While adding both these numbers, the latter will contribute the rightmost non-zero integer of the sum (Since, it has minimal number of 0s).
So, the number 2840^125(the value should be 284^125) should be considered.
The power cycle for 4 is 4, 6.
4^odd value is 4
4^even value is 6
Here the power is odd , 4^125.So the rightmost non-zero integer is 4.

25. The unit digit of the expression 68^24 * 25^45 * 93^21 is _____. **(AMAZON)**
A) 5 B) 0 C) 2 D) 3
Sol:5 power anything is going to end with 5
8 power anything is going to end with a even number
Even number * 5 is always going to end with a zero. Any number * 0 will definitely ends with a zero.
Hence, the unit digit is 0.

PRACTICE:

1. The number of prime numbers between 0 and 50 is
   (*a*) 14 (*b*) 15 (*c*) 16 (*d*) 17

2. Which one of the following is a prime number?
   (*a*) 161 (*b*) 221 (*c*) 373 (*d*) 437

3. The remainder obtained when any prime number greater than 6 is divided by 6 must be
   (*a*) either 1 or 2 (*b*) either 1 or 3 (*c*) either 1 or 5 (*d*) either 3 or 5

4. (800/64) × (1296/36) =?
   (*a*) 420 (*b*) 450 (*c*) 500 (*d*) 540

5. The unit's digit of 13^2003 is
   (*a*) 1 (*b*) 3 (*c*) 7 (*d*) 9

6. The digit in the unit's place of the number 127^99 is
   (*a*) 3 (*b*) 9 (*c*) 7 (*d*) 1

7. The value of 5^2 + 6^2 + ... +10^2 + 20^2 is
   (*a*) 755 (*b*) 760 (*c*) 765 (d)780

8. The unit's digit in the product 274 × 318 × 577 × 313 is
   (*a*) 2 (*b*) 3 (*c*) 4 (*d*) 5

9. Unit's digit in (784)126 + (784)127 is
   (*a*) 0 (*b*) 4 (*c*) 6 (*d*) 8

10. Which of the following numbers is not divisible by 18?
    (*a*) 34056 (*b*) 50436 (*c*) 54036 (*d*) 65043

11. Which one of the following numbers is divisible by 11?
    (*a*) 4823718 (*b*) 4832718 (*c*) 8423718 (*d*) 8432718

12. If 78*3945 is divisible by 11, where * is a digit, then * is equal to
    (*a*) 0 (*b*) 1 (*c*) 3 (*d*) 5

13. The difference of a number consisting of two digits and the number formed by interchanging the digits is always divisible by
    (*a*) 5 (*b*) 7 (*c*) 9 (*d*) 11

14. The difference between the squares of two consecutive odd integers is always divisible by
    (*a*) 3 (*b*) 6 (*c*) 7 (*d*) 8

15. What least number must be added to 1056 to get a number exactly divisible by 23?
    (*a*) 2 (*b*) 3 (*c*) 21 (*d*) 25

16. Which is the greatest 5-digit number exactly divisible by 279?
    (*a*) 99603 (*b*) 99550 (*c*) 99882 (*d*) None of these

17. The sum of 4 Consecutive even numbers is 748 ,The smallest number is
(a)185 (b)184 (c)186 (d)188

18. What is the difference between the biggest and the smallest fraction among 2/ 3 ,3 /4 , 4/ 5 and 5/6
(a)1/6 (b)1/12 (c)1/20 (d)3/20

19. 5.5 – [6.5 – {3.5 /(6.5 – 5.5 -2.5)}] is equal to
(*a*) – 1 (*b*) 0 (*c*) 0.1 (*d*) 1

20. Solve ?/529=324/?
(*a*) 404 (*b*) 408 (*c*) 410 (*d*) 414

21. 999 × 99 × 9 ÷ 99 ÷ 9 ÷ 3 =?
(*a*) 99 (*b*) 111 (*c*) 333 (*d*) 999

22. Find The Square root of 2304
(a)48 (b)49 (c)42 (d)46

23. Find The Cube root of 2197
(a)12 (b)11 (c)13 (d)15

24. Cost of 8 pens and 4 pencils is ` 176 and the cost of 2 pens and 2 pencils is ` 48. What is the cost of one pen?
(*a*) 12 (*b*) 14 (*c*) 16 (*d*) 18

25. $\sqrt{(\sqrt{17956}+\sqrt{24025})}$=?
(*a*) 19 (*b*) 155 (*c*) 256 (*d*)17

26. What is the smallest number to be subtracted from 549162 in order to make it a perfect square? **(TCS)**
(*a*) 28 (*b*) 36 (*c*) 62 (*d*) 81

27. How much % of numbers from 1-100 are composite numbers **(WIPRO)**
(a)74% (b)76% (c)75% (d)77%

28. What is the probability of selecting a prime number from 1-50 **(COGENT)**
(a)3/10 (b)1/5 (c)2 (d)2/5

29. How many digits are required to print the page numbers of a book having 200 pages. **(CTS)**
(a)482 (b)492 (c)502 (d)532

30. How many prime numbers in between 1-500 **(ACCENTURE)**
(a)95 (b)73 (c)75 (d)100

KEY:

1. B	11. B	21.C
2. C	12. D	22.A
3. C	13. C	23.C
4. B	14. D	24.C
5. C	15. A	25.D
6. A	16. A	26.D
7. A	17. B	27.A
8. A	18. A	28.A
9. A	19. B	29.B
10. D	20. D	30.A